ADAPTIVE-RESONANCE-TRACKED PHONONIC PIEZOELECTRIC LIGHT SOURCE

Extended Computational Monograph

From 100 phononic metacrystals to D067-ART-v1 adaptive resonance tracking

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Status: Reduced-order architecture closed at D067-ART-v1. Passive fixed-frequency operation rejected; adaptive resonance tracking frozen into the concept. Full 3-D tensor FEM and experiment remain outstanding.

Abstract

This monograph documents a local-compute design program that began with a 100-phononic-metacrystal Bloch-spectrum screen and progressively converted the most promising resonant families into a finite electromechanical light-source model. The work used a sequence of increasingly restrictive models: periodic anti-plane shear screening, finite bulb-scale reduced-order dynamics, independent zero-dependency verification, five-dimensional electromechanical optimization, physically linked piezoelectric closure, manufacturing-corner analysis, and robust-control reformulation. The principal outcome is not a claim of illumination-class performance. It is the identification of a coherent architecture, D067-ART-v1, in which a nested inertial phononic resonator couples to a PZT-5A transducer and requires active resonance tracking because the useful passive resonance is too narrow to survive ordinary manufacturing detuning at fixed frequency and load.

The physically constrained reference point occurs near 1.83592556468478932e+02 Hz with load resistance 5.29606372893091338e+03 ohm. At a modeled peak base acceleration of 1.00000000000000000 m/s^2, the reduced-order model gives piezoelectric AC power 1.29380171299124175e-04 W and optical power 4.42790698254122647e-05 W after the assumed rectification, conversion, and LED chain. Nominal shell and piezoelectric safety factors are 4.47813800228649495 and 2.01643731830532191. The resonance has half-power bandwidth 3.06168876413914859 Hz and effective Q = 5.99644740572118309e+01, which explains the failure of passive fixed-frequency robustness under +/- manufacturing variation.

> **Scope discipline:** Every numerical value in this document belongs to a reduced-order model unless explicitly labeled otherwise. No experimental validation, illumination-class performance, fatigue lifetime, thermal certification, or full 3-D tensor piezoelectric FEM is claimed.

Contents

1. Program Objective and Model Hierarchy

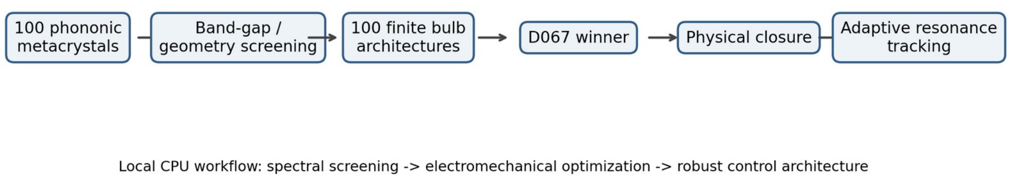
From spectral screening to a closed control architecture

The original computational objective was deliberately broad: generate and solve a large family of phononic metacrystals locally, identify structures with useful band-structure or localization behavior, and then determine whether that information could support a finite piezoelectric light-source concept. The first solver was therefore a screening instrument rather than a final device model.

The study evolved through a sequence of falsification gates. Each gate either preserved a mechanism under stronger constraints or removed an artifact introduced by the previous model. This progression matters because several attractive intermediate optima were later rejected or reinterpreted when the model was widened or when previously independent parameters were physically linked.

The final result is architectural. A nested inertial resonator remains useful, PZT-5A survives the physical closure stage, and the operating resonance is identifiable. The passive production architecture does not survive manufacturing variation at one fixed operating frequency and load. Adaptive tracking is therefore not an optional enhancement; it is part of the frozen D067-ART-v1 concept.

- Stage 1: 100-design phononic spectral screening.
- Stage 2: finite bulb electromechanical screening.
- Stage 3: independent zero-dependency audit.
- Stage 4: broadband and five-dimensional optimization.
- Stage 5: physical coupling/capacitance closure.
- Stage 6: manufacturing-corner robustness.
- Stage 7: adaptive-resonance architecture freeze.



Program evolution from phononic screen to adaptive light-source architecture.

2. Computational Environment and Local-First Workflow

Why the remote droplet became unnecessary for this model class

The 100-metacrystal and subsequent reduced-order bulb studies were executed on local CPU resources. The computational form was well matched to sparse eigenproblems, small dense reduced-order systems, deterministic parameter sweeps, and multiprocessing across independent designs. These workloads did not require remote infrastructure once the local solver architecture was stabilized.

The local-first workflow also improved iteration latency. Geometry changes, new constraints, and independent checks could be executed immediately without provisioning cloud resources. The computational ledgers later reached millions of coupled solves while remaining practical because the reduced-order response at each parameter point was analytically condensed to a small complex system.

This should not be generalized to the next stage. A full 3-D tensor piezoelectric finite-element model with fine geometry, contacts, electrodes, and thermal coupling can become memory- and factorization-limited. The local-first conclusion applies to the screening and reduced-order phases documented here.

> **Compute boundary:** Local CPU compute is sufficient for broad reduced-order screening. Cloud or GPU resources become materially useful again for high-fidelity 3-D multiphysics FEM, large uncertainty ensembles, or transient nonlinear models.

3. Phononic Spectral Formulation

Periodic anti-plane shear screening

The initial phononic solver used the anti-plane shear sector of linear elastodynamics. This is a legitimate scalar sector for out-of-plane displacement in an appropriate 2-D configuration, but it is not equivalent to the full in-plane elastic tensor problem.

Spatial variation of shear modulus and density encoded the metacrystal geometry. Bloch periodicity converted the unit-cell problem into a wave-vector-dependent Hermitian generalized eigenproblem. The solver sampled the conventional square-lattice high-symmetry path Gamma-X-M-Gamma.

Band gaps detected only along a high-symmetry path were labeled path gaps. A separate full-Brillouin-zone sampling stage was used for stronger verification of selected candidates because a path gap does not automatically prove a complete 2-D gap.

$$-\mathrm{div}[\ \mathrm{mu}(r)\ \mathrm{grad}\ u_z(r)\ ] = \mathrm{rho}(r)\ \mathrm{omega}^2\ u_z(r)$$

$$u_z(r + R) = \exp(i\ k.R)\ u_z(r)$$

$$\mathrm{Gamma} \to X \to M \to \mathrm{Gamma}$$

4. One-Hundred-Metacrystal Design Space

Ten geometry families times ten parameter states

The revised spectral solver did not merely vary two radii in one geometry. It enumerated ten structurally distinct geometry families, each with ten parameter states, producing one hundred named designs. Material assignments and geometric parameters were embedded in the manifest so that numerical identifiers such as D000-D099 remained traceability tags rather than substitutes for engineering names.

The families were selected to span locally resonant, anisotropic, clustered, annular, split-ring, dimerized, and labyrinthine motifs. The purpose was not to assert that every family is fabrication-optimal. The purpose was to create a structurally heterogeneous screening set large enough to expose whether particular mechanisms repeatedly survived later model changes.

Family	Representative structural concept

Circular core-shell	Dense core, compliant shell, elastic matrix
Square core-shell	Square nested local resonator
Elliptical core-shell	Anisotropic local resonance
Cross resonator	Cruciform inertial inclusion
Four-cylinder cluster	Clustered multiple inclusions
Annular resonator	Ring-localized resonance
Split-ring	Broken rotational symmetry
Rotated diamond	Oriented anisotropic inclusion
Dimer resonator	Paired local resonators
Orthogonal labyrinth	Long compliant path / compact inertial mass

5. Bloch Solver Numerical Architecture

Sparse complex Hermitian eigenproblems

Each rasterized unit cell produced spatial arrays for density and shear modulus. Periodic finite differences with Bloch phase factors enforced the wave-vector-dependent boundary conditions. Harmonic averaging across material interfaces improved flux consistency in the discrete operator.

The CPU implementation used sparse eigensolvers to extract the lowest branches. Independent metacrystal designs were parallelized at the process level while BLAS threading was restricted to prevent oversubscription. Progress reporting was moved to the Bloch-point level so the user could verify that the solver was genuinely traversing reciprocal space rather than appearing stalled.

Outputs included a design manifest, per-design band CSVs, partial and final summary tables, plots, and checkpoint files. This output structure later became the template for the bulb-scale optimizer.

$$A(k)\, u = \omega^2 M u$$
$$N_{DOF} = N_x N_y$$
$$N_{primary} = N_{design} N_k$$

6. Coarse Spectral Results and Brillouin-Zone Verification

Screen first, verify second

The initial 100-design spectral run completed all designs and identified high relative path gaps in several families. A heavier run increased spatial resolution, number of bands, and number of k-points. The winner changed between coarse and heavier runs, demonstrating that the coarse result was correctly functioning as a screening stage rather than a converged final ranking.

Full two-dimensional Brillouin-zone verification was then applied to selected top candidates. This distinction is essential: the complete flag belongs only to the sampled 2-D verification, not to a Gamma-X-M-Gamma path gap. The computational workflow therefore separated inexpensive broad ranking from more expensive completeness checks.

> **Interpretation rule:** Rank reversals under refinement are not automatically numerical failure. They are evidence that the coarse stage is under-resolved and must be treated only as a candidate generator.

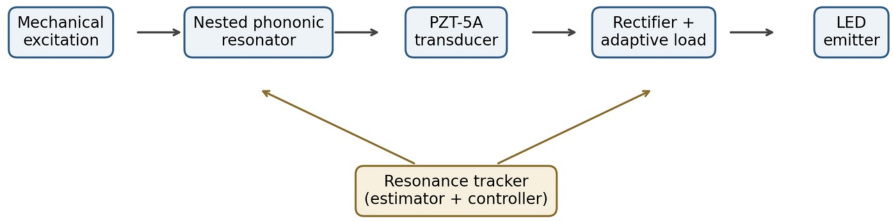
7. Transition from Infinite Crystal to Finite Bulb

Changing the objective from band gap to energy conversion

The light-source problem is not solved by maximizing a band gap. A finite device must place mechanical strain where piezoelectric material can convert it to electrical energy. The design objective therefore moved from spectral exclusion to electromechanical concentration, electrical loading, stress control, and optical output.

The bulb-scale model replaced the infinite periodic lattice with a finite circumferential sector network. Each sector carried shell motion, local inertial motion, and a piezoelectric electrical channel. The architecture therefore admitted finite resonances, mode localization, load optimization, and explicit power ledgers.

The intended physical stack became: external or self-driven mechanical excitation; nested phononic resonator; piezoelectric extraction; rectification and DC conversion; optical emitter. This is a substantially different optimization problem from the original Bloch-band screen.



Conceptual transition to finite device and adaptive resonance tracking.

8. Coupled Electromechanical Reduced-Order Model

Mechanical dynamics plus piezoelectric electrical back-action

For harmonic motion, the mechanical and electrical subsystems were represented by a coupled linear system. The piezoelectric voltage was not calculated after the fact from a mechanically uncoupled solution. Instead, the electrical subsystem was condensed into the mechanical dynamic stiffness so that electrical loading modified the mechanical response.

This approach preserved the key reduced-order feedback mechanism: voltage generation depends on motion, load current depends on voltage, and the electrical admittance feeds back into the effective dynamic stiffness through the coupling matrix.

The model remains a reduced-order representation. It does not resolve three-dimensional tensor fields, electrode edge effects, adhesive layers, nonlinear hysteresis, ferroelectric domain dynamics, or thermal gradients.

(K - omega^2 M + i omega C) u - Theta V = F
i omega Theta^T u + Y V = 0
Y = 1/R_L + i omega C_p
[D_m + i omega Theta Y^(-1) Theta^T] u = F

9. The First 100-Bulb Sweep

Finite architecture ranking across frequency and load

The bulb solver enumerated one hundred finite resonant architectures across ten families. Each design was swept over frequency and load resistance. The model computed piezoelectric AC power, rectified power, LED electrical power, optical power, mechanical dissipation, stress proxies, peak voltage, efficiency, and a broadband score.

The first broad sweep was intentionally permissive. It was designed to reveal which finite architectures generated coherent resonance and useful coupling before the physical relation between capacitance and coupling was fully enforced. D067 repeatedly emerged as a strong working family and was promoted for dedicated audits.

Bulb family	Design intent
Uniform Annular Core-Shell Resonator	finite resonant architecture
Dimerized Heavy-Light Resonator	finite resonant architecture
Quadrupolar Cluster Resonator	finite resonant architecture
Split-Ring Defect Resonator	finite resonant architecture
Elliptically Graded Resonator	finite resonant architecture
Labyrinthine Compliant-Path Resonator	finite resonant architecture
Nested Inertial Core-Shell Resonator	finite resonant architecture
Fibonacci Binary-Mass Resonator	finite resonant architecture
Azimuthally Graded Resonator	finite resonant architecture
Localized Defect-Cavity Resonator	finite resonant architecture

10. D067 Selection and Interpretation

Nested inertial core-shell resonator architecture

D067 corresponds to a nested inertial core-shell resonator bulb architecture. The family combines a structural shell, dense local inertial mass, compliant coupling, and circumferential piezoelectric participation. It became the working architecture because it continued to produce strong resonant response after the model was widened and because its behavior could be interpreted in terms of identifiable lower and upper mechanical branches.

At this stage D067 was a research branch, not a frozen hardware design. Several subsequent calculations changed the predicted optimum substantially. Those changes are documented rather than hidden because the sequence is itself evidence about which numerical claims are robust and which are artifacts of model assumptions.

11. Zero-Dependency Audit

Independent standard-library calculation

A self-contained two-degree-of-freedom audit was implemented without NumPy, SciPy, pandas, plotting libraries, or imports from the main solver. The audit used analytic inversion of a complex 2x2 dynamic system and therefore served as a structurally independent check on resonance, load matching, power scaling, and stress proxies.

The first audit swept 1001 frequencies by 121 electrical loads, producing 121121 primary coupled evaluations plus a fixed-load bandwidth scan. It immediately showed that an apparent optimum near 501.928938216837707 Hz was not an interior optimum: optical power was still rising at the upper sweep boundary near 752.893407325256590 Hz.

The electrical load, however, behaved correctly. The optimum load was close to the capacitive reactance, with omega R C approximately equal to one. This separated an electrical result that was behaving sensibly from a mechanical frequency window that was too narrow.

omega R C_p ~= 1
P_load = 0.5 |V|^2 / R_L

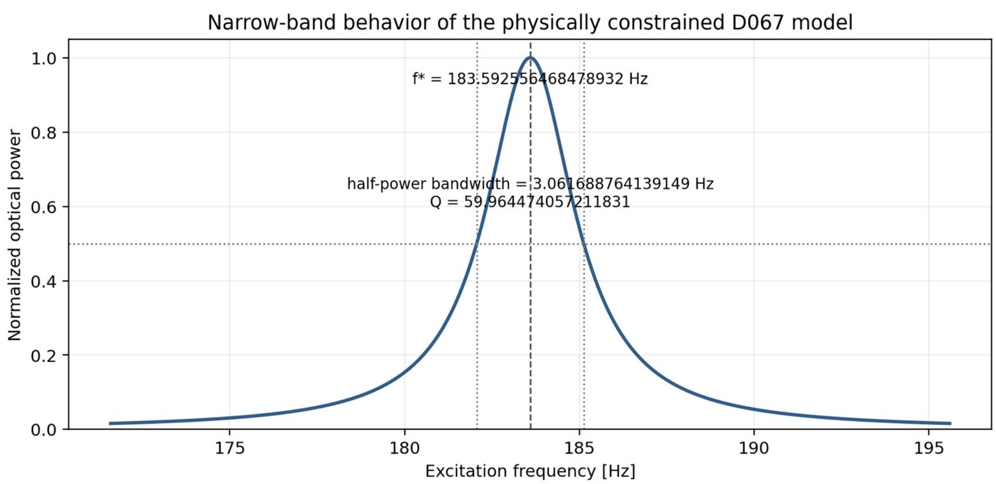
12. Broadband Resonance Correction

The useful lower branch is near 156.659 Hz in the nominal audit

The next independent broadband calculation extended the frequency window to 50-3000 Hz and derived the two undamped modal branches analytically. The nominal two-degree-of-freedom model contained mechanical modes near 1.56659049596398518e+02 Hz and 7.80045024052060285e+02 Hz.

The broadband baseline optimum moved to 1.56630606946179341e+02 Hz with optical output 3.89707226331559452e-05 W. This established that the previously observed high-frequency edge growth belonged to the upper branch, whereas the stronger useful response for the nominal system was concentrated around the lower mode.

This correction is a key example of why wide frequency windows and interior-optimum checks are mandatory for resonant design studies.



Representative resonance-focused diagnostic used in the report.

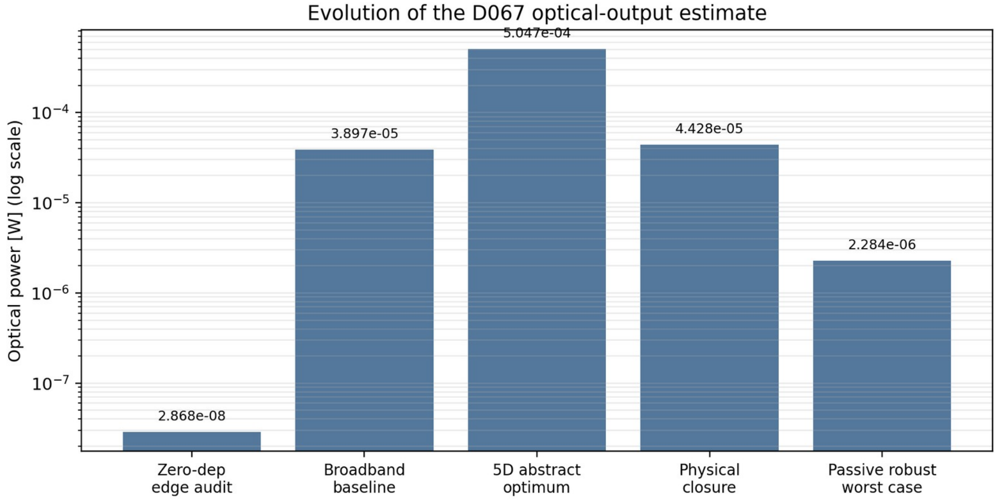
13. Five-Dimensional Exploratory Optimization

Performance direction before enforcing piezo geometry closure

A deterministic 1024-architecture search varied five reduced parameters: electromechanical coupling scale, resonator mass scale, local stiffness scale, capacitance scale, and damping scale. This search was intentionally exploratory. It asked which direction in abstract parameter space increased optical power while respecting shell, piezo, and voltage constraints.

The best constrained abstract point used theta scale 2.0, mass scale 1.5, local stiffness scale 1.5, capacitance scale 0.5, and nominal damping scale. It predicted optical power 5.04745284881485067e-04 W and a 12.9519098127282923-fold gain over the baseline reduced-order D067 configuration.

That point was not frozen because theta and C_p had been varied independently. In a realizable piezoelectric element, material constants and geometry couple these parameters. The result was retained as a direction-of-improvement signal, not as a physical design claim.



Evolution of modeled power as increasingly restrictive constraints were imposed.

14. Physical Closure of Coupling and Capacitance

Removing an artificial degree of freedom

The next solver enforced a physical relation between electrode geometry, dielectric permittivity, effective electromechanical coupling, capacitance, and the structural stiffness scale. Piezoelectric material, electrode coverage, thickness, active height, resonator mass, local stiffness, and damping were searched together.

Capacitance was calculated from electrode geometry and dielectric permittivity. Effective coupling was derived from a material coupling measure multiplied by a modal-overlap factor. Theta was then constructed from the same capacitance and structural stiffness scale. This removed the ability to obtain artificial performance by independently increasing coupling while decreasing capacitance.

The search included PZT-5H, PZT-5A, AlN, and LiNbO3 and required nominal shell and piezoelectric safety factors of at least 2.0.

C_p = epsilon_33 A_p / t_p
gamma_material^2 = e31^2 / (E_p epsilon_33)
kappa_eff^2 = overlap^2 gamma_material^2
Theta = sqrt(kappa_eff^2 K_eff C_p)

15. Thirty-Thousand-Candidate Constrained Search

Physical parameter search with 16.47 million primary solves

The physical-closure grid contained 30000 parameter combinations. Each candidate was evaluated over an adaptive frequency set and an electrical load set. The computational ledger for the primary stage reached 16470000 coupled solves, followed by robustness and dense local refinement.

The strongest nominal physically linked candidate was PZT-5A. The robust winner among the top audited designs used coverage 0.65, piezo thickness 6.5e-4 m, active-height fraction 0.9, resonator mass scale 1.0, local stiffness scale 1.75, and damping scale 1.25.

The final dense refinement placed the operating point essentially on the lower mechanical mode: mode 1 = 1.83592556468482883e+02 Hz and refined operating frequency = 1.83592556468478932e+02 Hz. The numerical agreement is expected because the reduced-order model is sharply resonant.

Quantity	Physically constrained D067 value
Piezo material	PZT-5A
Coverage	6.50000000000000022e-01
Piezo thickness	6.49999999999999970e-04 m
Active height fraction	9.00000000000000022e-01
Mass scale	1.00000000000000000e+00
Local stiffness scale	1.75000000000000000e+00
Damping scale	1.25000000000000000e+00
Mode 1	1.83592556468482883e+02 Hz
Mode 2	7.53481662427537117e+02 Hz

16. Robustness and Manufacturing Corners

Why nominal success is insufficient

The physically linked nominal design passed the shell, piezoelectric, voltage, coupling, and interior-frequency checks. It failed the chosen robustness criterion. Thirty-two deterministic manufacturing corners perturbed resonator mass, coupling stiffness, piezo thickness, electrode coverage, and damping.

The robust physical winner had nominal optical power 4.42790698254155377e-05 W, but the worst manufacturing corner after corner-specific retuning produced 1.98685301570967799e-06 W. The robustness ratio was only 4.48711552330138055e-02. This was a decisive rejection of a naive passive fixed-geometry interpretation.

A later fixed-frequency all-design robustness screen produced a best worst-case value 2.28401432071939221e-06 W, still only a small fraction of nominal output. The result is consistent with a narrow resonant system whose manufacturing-induced detuning exceeds the usable resonance width.

$$R\_robust = P\_worst / P\_nominal$$
$$R\_robust = 4.48711552330138055e-02$$

17. Why Passive Fixed-Frequency Operation Fails

A narrow resonance meets finite manufacturing tolerances

The physically constrained half-power bandwidth is 3.06168876413914859 Hz around an operating frequency near 183.59255646848 Hz. The corresponding effective Q is 5.99644740572118309e+01, giving a fractional half-power bandwidth of about 1.6676540830590867e-02.

Manufacturing variations of several percent in mass or stiffness can shift resonance by an amount larger than this usable bandwidth. A fixed-frequency, fixed-load device can therefore move from the top of the resonance to a low-power shoulder even while all material strengths remain acceptable.

A robustness ratio by itself is not a sufficient objective. An optimizer can achieve a high ratio by moving far from resonance until every manufacturing corner produces nearly the same negligible power. Robust design must preserve both an absolute power floor and acceptable retention of nominal output.

> **Rejected architecture:** A passive fixed-frequency, fixed-load production bulb is rejected by the reduced-order robustness study.

18. Adaptive Resonance Tracking Architecture

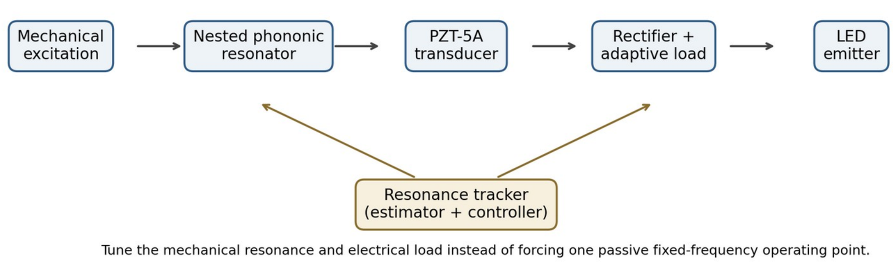
D067-ART-v1

The appropriate response to detuning is not necessarily to destroy the resonance by adding large passive damping. A resonant converter exists because concentration of mechanical energy is useful. The preferred architecture retains the resonance and makes the operating point adaptive.

D067-ART-v1 therefore combines the nested inertial resonator, PZT-5A transducer, resonance estimator, tuning element or controlled drive frequency, and adaptive electrical load. For a self-driven device, the actuator frequency can track the measured resonance. For externally imposed vibration, effective stiffness, effective mass, or an electrical shunt must be adjusted so that device resonance follows the excitation.

The control objective should maximize absolute electrical or optical power subject to stress, voltage, temperature, and actuator constraints. Robustness ratios can be monitored but must not become the sole objective.

$$f\_r = (1 / 2 pi) sqrt(k\_eff / m\_eff)$$
$$e\_f = f\_excitation - f\_r$$
$$R\_L*(omega) \sim 1 / (omega C\_p)$$



Adaptive resonance tracking concept frozen into D067-ART-v1.

19. Electrical Load Tracking

Impedance matching is a second control loop

The independent audits repeatedly produced load optima close to the expected RC scale. In a simple piezoelectric load model, the useful resistive load remains on the order of the capacitive reactance. Because the resonant frequency can move with manufacturing tolerance, temperature, preload, and aging, the useful load also moves.

The device therefore benefits from two coordinated adaptive variables: mechanical resonance alignment and electrical impedance alignment. A practical implementation could switch among discrete resistor values, use a controlled converter input impedance, or implement synchronous energy extraction. The reduced-order model only represents a resistive load, so converter topology optimization remains a later task.

X_C = 1 / (omega C_p)
omega R_L C_p ~ O(1)

20. Stress, Power, and Photon Ledgers

The physically constrained reference point

At the final physically constrained operating point and 1.00000000000000000 m/s^2 peak base acceleration, piezoelectric AC power is 1.29380171299124175e-04 W and modeled optical power after the assumed conversion chain is 4.42790698254122647e-05 W.

The corresponding mechanical-to-electrical efficiency is 1.83361290460046060e-02 and total optical efficiency is 6.27535680470461734e-03. Peak modeled voltage is 1.17064566155618577 V. Shell and piezoelectric safety factors are 4.47813800228649495 and 2.01643731830532191.

At a 555 nm reference wavelength, the optical power corresponds to a photon rate 1.23712829446936031e+14 s^-1 and an idealized monochromatic photopic equivalent 3.02426046907565763e-02 lumen. The latter is a reference conversion, not a prediction of actual broadband LED luminous efficacy.

Metric	Value
Operating frequency	1.83592556468478932e+02 Hz
Load resistance	5.29606372893091338e+03 ohm
Piezo AC power	1.29380171299124175e-04 W
Optical power	4.42790698254122647e-05 W
Mech->elec efficiency	1.83361290460046060e-02
Total optical efficiency	6.27535680470461734e-03
Peak voltage	1.17064566155618577 V
Shell SF	4.47813800228649495
Piezo SF	2.01643731830532191
Photon rate @555 nm	1.23712829446936031e+14 s^-1

21. Scaling and Performance Limits

Linear-response scaling is informative but not a hardware guarantee

Within the linear model, displacement and voltage scale linearly with base acceleration while power scales quadratically. This makes it possible to estimate where nominal stress or voltage ceilings would be reached. However, the extrapolation must stop before nonlinear elasticity, adhesive slip, ferroelectric nonlinearity, geometric stiffening, impact, or fatigue invalidates the linear model.

For the physically constrained design, the piezoelectric stress criterion is reached before the voltage ceiling. At the selected safety-factor-2 design criterion, the available acceleration margin is only slightly above the 1 m/s^2 reference point. This makes higher output through simple excitation scaling unattractive without a geometry redesign that increases useful electrical coupling per unit peak piezoelectric stress.

The more promising route is therefore modal strain redistribution, larger active volume, multiple transducers, multi-resonator staggering, or improved transduction and conversion electronics rather than brute-force acceleration.

V ~ a
P ~ a^2

22. Control-System Realization Options

How adaptive tracking could be implemented

An adaptive resonance tracker requires a measurable state correlated with resonant alignment. Candidate observables include piezoelectric voltage amplitude, electrical power, phase between excitation and response, motional impedance, or a dedicated strain/acceleration sensor. A hill-climbing controller can maximize output directly, while a phase-locked approach can track a known resonance condition more rapidly.

For a self-driven device, the actuator frequency is the simplest tuning variable. For a vibration-harvesting device, resonance must instead be moved. Options include variable preload, magnetically biased stiffness, switchable shunt networks, mechanically adjustable inertial mass, or a secondary actuator that perturbs effective stiffness.

Control bandwidth should be chosen relative to the expected rate of environmental or thermal drift, not relative to the 183.6 Hz carrier itself. The resonance can be tracked slowly while the mechanical system continues oscillating rapidly.

- Direct power hill-climb: perturb frequency or tuning state and retain changes that increase electrical power.
- Phase tracking: regulate phase toward a calibrated resonant value.
- Impedance tracking: estimate motional impedance and align converter input impedance.
- Hybrid method: phase-lock mechanically, then optimize electrical load locally.

23. Mechanical Realization Options

Turning the nested resonator abstraction into hardware

The reduced-order D067 model contains a shell generalized mass, a dense local inertial mass, and a compliant coupling. A hardware design must map these quantities to manufacturable geometry while preserving the target lower resonance and maintaining stress margins. Tungsten remains attractive for compact inertial mass because of its density, while polymer or elastomer elements provide compliant coupling.

Practical architectures include concentric ring masses suspended by flexural spokes, segmented annular masses coupled through elastomer webs, or nested axial/circumferential frames. The detailed geometry should be designed by matching modal mass and stiffness rather than copying the visual concept image.

The transition to finite-element modeling should therefore preserve the reduced-order targets: modal frequency, modal mass participation, strain concentration in the piezoelectric region, and separation from the upper mechanical branch.

> **Geometry principle:** Match modal quantities, not appearances. The reduced-order D067 parameters define target dynamics, not a literal CAD drawing.

24. Piezoelectric Material Comparison

Why PZT-5A became the reduced-order winner

The physically constrained search included PZT-5H, PZT-5A, AlN, and LiNbO3. PZT-5A dominated the top physically linked designs in the selected search grid. The result reflects the interaction of electromechanical coefficient, dielectric permittivity, stiffness, mass, allowed stress, electrode geometry, and the assumed overlap factor.

PZT-5H can provide high coupling but also high dielectric capacitance and different stress characteristics. AlN and LiNbO3 offer attractive stiffness, stability, and potentially different fabrication pathways, but the current reduced-order bulk-scale parameterization did not select them for maximum optical output under the imposed constraints.

This ranking is model-dependent. Material selection must be revisited with actual vendor data, geometry-specific tensor orientation, temperature limits, fatigue data, and electrode construction.

Material	Role in current screen	Caution
PZT-5A	Selected physical winner	Use measured tensor constants and stress/fatigue data
PZT-5H	Strong exploratory coupling	High permittivity changes electrical loading
AlN	High stiffness, low dielectric constant	Thin-film fabrication context differs from bulk PZT
LiNbO3	Low dielectric loss / crystalline option	Orientation and tensor coupling must be modeled explicitly

25. Uncertainty and Sensitivity

What the model has already shown to be sensitive

The study exposed sensitivity in frequency window, spatial resolution, parameter coupling, manufacturing variation, and objective-function definition. Each sensitivity changed the interpretation of an apparent optimum. These are not secondary details; they define the credibility of the reduced-order conclusion.

The largest practical sensitivity is resonance detuning. The narrow half-power bandwidth means that modest errors in effective mass or stiffness dominate fixed-frequency output. The second major sensitivity is the relation between capacitance and coupling. Treating them as independent produced an optimistic abstract optimum that did not survive physical closure.

Future uncertainty propagation should sample measured distributions rather than symmetric engineering corners once prototype statistics exist.

- Frequency-window truncation can create false boundary optima.
- Mesh refinement can change spectral rankings.
- Independent theta and C_p scaling can produce nonphysical gains.

- Fixed-frequency output is highly sensitive to mass and stiffness tolerances.
- Robustness ratios without absolute power floors can reward uniformly bad designs.

26. Validation Hierarchy

A sequence of increasingly strong claims

The project should be interpreted through a validation hierarchy. At the lowest level, algebraic and dimensional checks verify implementation consistency. At the next level, independent reduced-order formulations verify resonance and load behavior. Higher levels require finite-element convergence, tensor material models, experimental modal testing, and finally optical power measurements.

The present study closes only the reduced-order architecture level. It provides a candidate architecture and numerical targets for the next model. It does not close full-field mechanics or experimental performance.

Level	Validation object	Status
0	Algebra, units, solver execution	Completed
1	Independent zero-dependency ROM	Completed
2	Physically linked ROM optimization	Completed
3	Manufacturing-corner ROM	Completed; passive fixed-frequency rejected
4	Finite-sector tensor FEM	Not completed
5	3-D piezoelectric/thermal FEM	Not completed
6	Bench modal/electrical prototype	Not completed
7	Optical output / lifetime validation	Not completed

27. Full-FEM Escalation Plan

What the next serious simulation must contain

The next meaningful numerical escalation is a finite-sector or full three-dimensional piezoelectric model. The displacement field should include all three components. The constitutive relation must use the anisotropic elastic stiffness tensor, piezoelectric coupling tensor, and dielectric tensor in the correct material orientation.

Electrodes should be modeled as equipotential surfaces or conductive domains with explicit circuit boundary conditions. Adhesive layers, compliant coupling elements, contact/preload, and realistic supports should be included because each can move the narrow resonance substantially. Modal and harmonic-response solutions should be compared with the reduced-order target before any transient calculation is attempted.

Mesh convergence should be demonstrated for resonance frequency, peak piezoelectric stress, stored energy, and electrical power. A converged FEM result that fails to reproduce the reduced-order mode near 183.6 Hz would trigger a redesign rather than a parameter adjustment.

div(sigma) + rho omega^2 u = 0
sigma = C:Epsilon - e^T E
D = e:Epsilon + epsilon E
E = -grad(phi)

28. Experimental Prototype Plan

A falsifiable bench test

A first prototype does not need to produce useful illumination. Its purpose is to determine whether the predicted mechanical-electrical coupling and narrow resonance exist at approximately the expected scale. The experiment should therefore prioritize clean excitation, modal identification, voltage/current measurement, and energy accounting.

A shaker or controlled actuator can provide sinusoidal base acceleration. Accelerometers measure the imposed motion. The piezoelectric output is measured across a selectable resistive or programmable load. A frequency sweep maps mechanical and electrical response simultaneously. The resonance-tracking algorithm can then be tested by intentionally adding small masses or changing preload to simulate manufacturing detuning.

The optical stage should initially be replaced by an electrical load and power meter. Only after the electrical ledger closes should a rectifier/DC converter and LED be added. This ordering prevents optical-conversion uncertainty from hiding mechanical or piezoelectric errors.

- Measure base acceleration and structural response simultaneously.
- Sweep frequency through both predicted mechanical branches.
- Measure open-circuit voltage and loaded power across multiple R_L values.
- Extract Q and compare with reduced-order and FEM predictions.
- Perturb mass/stiffness deliberately and test adaptive tracking recovery.

29. Falsification Criteria

Conditions that would reject the present architecture

The architecture should be abandoned or substantially revised if the next validation stage contradicts its central reduced-order predictions. A rigorous development program needs explicit rejection criteria rather than only improvement targets.

Examples include failure to observe a corresponding lower mechanical mode, inability to place useful strain in the piezoelectric region without exceeding stress limits, order-of-magnitude disagreement in electrical power after calibrated excitation, or adaptive tracking that cannot recover power under deliberate detuning.

A negative result at any of these gates is scientifically useful because it localizes the failure mechanism.

- No identifiable lower mode near the design target after geometry is calibrated.
- Piezoelectric stress reaches allowable limits before useful electrical output appears.
- Measured optimum load is incompatible with the modeled capacitance scale.
- Adaptive tuning cannot recover the resonant peak after controlled detuning.
- Thermal drift moves resonance faster than the controller can track.
- Full FEM predicts qualitatively different mode localization.

30. Conclusions and Frozen Status

What can be concluded now

The 100-design phononic screen, finite bulb optimizer, independent zero-dependency audit, broadband correction, physically linked piezoelectric closure, and robustness analysis converge on one defensible architectural statement: a nested inertial phononic-piezoelectric light-source concept can produce a coherent resonant electromechanical response in the reduced-order model, but the useful passive resonance is too narrow for fixed-frequency manufacturing robustness.

The working architecture is therefore frozen as D067-ART-v1: a nested inertial resonator with PZT-5A transduction, adaptive resonance tracking, and adaptive electrical loading. The physically constrained reference optical power is 4.42790698254122647e-05 W at 1 m/s^2 peak base acceleration under the assumed conversion chain.

The reduced-order program is closed at the architecture level. Further unconstrained parameter sweeps would have diminishing scientific value. The next informative work is full-field FEM and a bench prototype designed to falsify or validate the modal, electrical, stress, and tracking predictions.

> **Frozen status:** D067-ART-v1 is a reduced-order architecture freeze, not a hardware certification and not a claim of a household replacement light source.

Appendix A. Symbol and Quantity Dictionary

Reduced-order notation used throughout the study

Symbol	Meaning
u	Mechanical generalized displacement vector
V	Piezoelectric voltage vector
M	Mechanical mass matrix
K	Mechanical stiffness matrix
C	Mechanical damping matrix
Theta	Electromechanical coupling matrix or scalar
C_p	Piezoelectric capacitance
R_L	Electrical load resistance
Y	Electrical admittance
omega	Angular frequency
f	Frequency
rho	Mass density
mu	Shear modulus
k_c	Local compliant coupling stiffness
m_r	Local resonator mass

Q	Effective resonance quality factor
SF	Safety factor
P_piezo	AC electrical power delivered by the piezoelectric model
P_optical	Optical power after assumed rectifier/DC/LED efficiencies
kappa_eff^2	Reduced effective electromechanical coupling measure

All symbols belong to reduced-order or linearized continuum models. Their numerical meaning depends on the stage of the study. In particular, Theta in the exploratory 5-D model was an abstract scale factor, while the physical-closure stage linked Theta to material and geometric parameters.

Appendix B. Numerical Milestones

Chronology of the principal computational results

Stage	Computation	Interpretation
100 phononic designs	All completed	Spectral screening established local workflow
Heavy phononic rerun	100/100	Rank reversal demonstrated need for refinement
First bulb sweep	100/100	D067 family promoted
Zero-dependency audit	122122 evaluations	Boundary optimum identified as window artifact
Broadband baseline	73841 evaluations	Lower mode near 156.63 Hz recovered
5-D exploratory search	1977457 coupled solves	12.9519x abstract gain found
Physical closure primary	16470000 coupled solves	PZT-5A physically linked winner
Physical robustness	32 corners	Passive fixed-frequency robustness failed
Fixed-point all-design screen	990000 corner evaluations	Best worst-case fixed point 2.28401432071939221e-06 W
Final architecture	D067-ART-v1	Adaptive resonance tracking frozen

Appendix C. D067 Physical Reference Parameters

Current reduced-order engineering targets

Parameter	Value
Bulb radius	3.34444444444444439e-02 m
Active height	5.75000000000000000e-02 m
Shell thickness	1.77333333333333338e-03 m
Piezo material	PZT-5A
Piezo coverage	6.50000000000000022e-01
Piezo thickness	6.49999999999999970e-04 m
Piezo height fraction	9.00000000000000022e-01
Resonator mass scale	1.00000000000000000e+00
Local stiffness scale	1.75000000000000000e+00
Damping scale	1.25000000000000000e+00
Capacitance	1.63686122094644199e-07 F
Coupling Theta	1.20949631597844637e-02

Effective kappa^2	1.23001189186170844e-03
Mode 1	1.83592556468482883e+02 Hz
Mode 2	7.53481662427537117e+02 Hz

Appendix D. Manufacturing-Corner Definition

Deterministic uncertainty set used in robustness screening

Parameter	Perturbation
Resonator mass	+/- 5 %
Local coupling stiffness	+/- 5 %
Piezo thickness	+/- 5 %
Electrode coverage	+/- 3 %
Damping	+/- 10 %

The Cartesian product of the five two-sided perturbations creates 2^5 = 32 deterministic corners. This is not a probabilistic yield model. It is a structured worst-case screen intended to expose sensitivity before measured manufacturing distributions are available.

$$N_{corners} = 2^5 = 32$$

Appendix E. Algorithmic Pseudocode

Core computational workflows

Bloch screening

```
for design in 100 metacrystals:
  rasterize rho(x,y), mu(x,y)
  for k on Gamma-X-M-Gamma:
    assemble Bloch-periodic sparse operator
    extract lowest eigenvalues
  detect path gaps
rank designs; verify selected candidates over 2-D BZ
```

Bulb response

```
for finite bulb design:
  assemble M, K, C, Theta, Cp
  for frequency:
    for load resistance:
      condense electrical subsystem
      solve complex mechanical response
      reconstruct V and P_load
      compute stress and efficiency
rank by constrained optical objective
```

Adaptive control

```
measure excitation and piezo response
estimate resonance error or output gradient
adjust drive frequency / stiffness / mass / shunt
adjust electrical load near useful impedance scale
enforce stress, voltage, thermal constraints
repeat until output objective converges
```

Appendix F. Model Assumptions and Non-Claims

Boundaries that must remain explicit

- Linear elasticity and linear piezoelectric response are assumed in the reduced-order models.
- The shell and piezo stress metrics are proxies derived from generalized displacement; they are not local 3-D stress maxima.
- The optical chain uses assumed rectifier, DC conversion, and LED wall-plug efficiencies.
- The 555 nm luminous-flux conversion is an idealized monochromatic reference, not an actual lamp spectrum.
- The anti-plane Bloch solver does not replace full tensor elasticity.
- The finite bulb model does not include electrode parasitics, rectifier switching losses, thermal drift, fatigue, adhesive creep, or nonlinear contact.

- No experimental validation has yet been performed.

Appendix G. FEM Validation Checklist

Minimum conditions before promoting the concept beyond reduced-order status

- Use anisotropic elastic, piezoelectric, and dielectric tensors in the correct crystal/poling orientation.
- Resolve the PZT, electrodes, adhesive, compliant coupling, inertial mass, shell, and support interfaces.
- Perform mesh refinement until modal frequency and electrical power converge.
- Verify that mode 1 remains near the reduced-order target after realistic geometry is introduced.
- Compare modal strain-energy localization with the reduced-order assumption.
- Compute local peak stresses and compare with vendor fatigue limits, not only static allowables.
- Couple a realistic external circuit or impedance boundary.
- Perform harmonic sweeps wide enough to capture both mechanical branches.
- Perturb mass and stiffness in the FEM model to test adaptive tracking authority.
- Add thermal analysis after the electromechanical model closes.

Appendix H. Bench-Test Data Products

What should be recorded during the first prototype campaign

Data product	Required measurement
Base excitation	time, acceleration, phase reference
Mechanical response	accelerometer or laser vibrometer amplitude and phase
Electrical open circuit	piezo voltage vs frequency
Loaded response	voltage, current, real power vs R_L and frequency
Modal fit	f_n, Q, damping, effective modal mass
Stress proxy	strain gauge or calibrated FEM-derived estimate
Tracking test	detuning step, recovery time, recovered power
Thermal drift	resonance shift vs temperature
Optical stage	LED electrical input and optical output after electrical closure

Appendix I. Final Research Status

Frozen reduced-order statement

> **Architecture freeze:** D067-ART-v1 = nested inertial phononic resonator + PZT-5A transducer + adaptive resonance tracking + adaptive electrical load + rectifier/DC stage + optical emitter.

What is established within the present model: a coherent lower resonant branch, physically linked piezoelectric coupling and capacitance, a constrained PZT-5A optimum, stress-limited rather than voltage-limited operation, and severe passive manufacturing sensitivity caused by narrow bandwidth.

What remains open: full-field tensor mechanics, actual fabricated geometry, adhesive and contact behavior, nonlinear and fatigue limits, thermal closure, converter topology, controller implementation, and experimental optical output.

The next research action is therefore not another unconstrained parameter sweep. It is a finite-element and experimental attempt to validate or falsify the frozen architecture.

Reduced-order architecture status: CLOSED AT D067-ART-v1

James Lockwood

Appendix J. Complete 100-Metacrystal Screening Manifest

Exact scientific names generated by the named spectral solver

The following table records the one hundred screening designs used by the revised phononic solver. D000-D099 are traceability identifiers. Scientific names encode material pairing, topology, and normalized geometry. The table is included to make the broad design-space screen auditable without requiring the source file.

ID	Family	State	q	Scientific name
D000	circular_core_shell	0	0.000000	Tungsten-Rubber/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal

				[r_core/a=0.105000, t_shell/a=0.035000]
D001	circular_core_shell	1	0.111111	Steel-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.118333, t_shell/a=0.041667]
D002	circular_core_shell	2	0.222222	Tungsten-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.131667, t_shell/a=0.048333]
D003	circular_core_shell	3	0.333333	Steel-Rubber/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.145000, t_shell/a=0.055000]
D004	circular_core_shell	4	0.444444	Tungsten-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.158333, t_shell/a=0.061667]
D005	circular_core_shell	5	0.555556	Steel-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.171667, t_shell/a=0.068333]
D006	circular_core_shell	6	0.666667	Tungsten-Rubber/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.185000, t_shell/a=0.075000]
D007	circular_core_shell	7	0.777778	Steel-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.198333, t_shell/a=0.081667]
D008	circular_core_shell	8	0.888889	Tungsten-Silicone/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.211667, t_shell/a=0.088333]
D009	circular_core_shell	9	1.000000	Steel-Rubber/Epoxy Locally Resonant Circular Core-Shell Phononic Crystal [r_core/a=0.225000, t_shell/a=0.095000]
D010	square_core_shell	0	0.000000	Tungsten-Rubber/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.105000, t_shell/a=0.030000]
D011	square_core_shell	1	0.111111	Steel-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.116667, t_shell/a=0.036111]
D012	square_core_shell	2	0.222222	Tungsten-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.128333, t_shell/a=0.042222]
D013	square_core_shell	3	0.333333	Steel-Rubber/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.140000, t_shell/a=0.048333]
D014	square_core_shell	4	0.444444	Tungsten-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.151667, t_shell/a=0.054444]

D015	square_core_shell	5	0.555556	Steel-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.163333, t_shell/a=0.060556]
D016	square_core_shell	6	0.666667	Tungsten-Rubber/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.175000, t_shell/a=0.066667]
D017	square_core_shell	7	0.777778	Steel-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.186667, t_shell/a=0.072778]
D018	square_core_shell	8	0.888889	Tungsten-Silicone/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.198333, t_shell/a=0.078889]
D019	square_core_shell	9	1.000000	Steel-Rubber/Epoxy Locally Resonant Square Core-Shell Phononic Crystal [h_core/a=0.210000, t_shell/a=0.085000]
D020	elliptical_core_shell	0	0.000000	Tungsten-Rubber/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.125000, a_y/a=0.080000, t_shell/a=0.025000, theta=0.000000 rad]
D021	elliptical_core_shell	1	0.111111	Steel-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.136111, a_y/a=0.086111, t_shell/a=0.030556, theta=0.087266 rad]
D022	elliptical_core_shell	2	0.222222	Tungsten-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.147222, a_y/a=0.092222, t_shell/a=0.036111, theta=0.174533 rad]
D023	elliptical_core_shell	3	0.333333	Steel-Rubber/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.158333, a_y/a=0.098333, t_shell/a=0.041667, theta=0.261799 rad]
D024	elliptical_core_shell	4	0.444444	Tungsten-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.169444, a_y/a=0.104444, t_shell/a=0.047222, theta=0.349066 rad]
D025	elliptical_core_shell	5	0.555556	Steel-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.180556, a_y/a=0.110556, t_shell/a=0.052778, theta=0.436332 rad]
D026	elliptical_core_shell	6	0.666667	Tungsten-Rubber/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.191667, a_y/a=0.116667, t_shell/a=0.058333, theta=0.523599 rad]
D027	elliptical_core_shell	7	0.777778	Steel-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.202778, a_y/a=0.122778, t_shell/a=0.063889, theta=0.610865 rad]

D028	elliptical_core_shell	8	0.888889	Tungsten-Silicone/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.213889, a_y/a=0.128889, t_shell/a=0.069444, theta=0.698132 rad]
D029	elliptical_core_shell	9	1.000000	Steel-Rubber/Epoxy Anisotropic Elliptical Core-Shell Phononic Crystal [a_x/a=0.225000, a_y/a=0.135000, t_shell/a=0.075000, theta=0.785398 rad]
D030	cross_resonator	0	0.000000	Tungsten-Rubber/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.180000, h_arm/a=0.035000, t_halo/a=0.022000, theta=0.000000 rad]
D031	cross_resonator	1	0.111111	Steel-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.193333, h_arm/a=0.039444, t_halo/a=0.025111, theta=0.087266 rad]
D032	cross_resonator	2	0.222222	Tungsten-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.206667, h_arm/a=0.043889, t_halo/a=0.028222, theta=0.174533 rad]
D033	cross_resonator	3	0.333333	Steel-Rubber/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.220000, h_arm/a=0.048333, t_halo/a=0.031333, theta=0.261799 rad]
D034	cross_resonator	4	0.444444	Tungsten-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.233333, h_arm/a=0.052778, t_halo/a=0.034444, theta=0.349066 rad]
D035	cross_resonator	5	0.555556	Steel-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.246667, h_arm/a=0.057222, t_halo/a=0.037556, theta=0.436332 rad]
D036	cross_resonator	6	0.666667	Tungsten-Rubber/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.260000, h_arm/a=0.061667, t_halo/a=0.040667, theta=0.523599 rad]
D037	cross_resonator	7	0.777778	Steel-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.273333, h_arm/a=0.066111, t_halo/a=0.043778, theta=0.610865 rad]
D038	cross_resonator	8	0.888889	Tungsten-Silicone/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.286667, h_arm/a=0.070556, t_halo/a=0.046889, theta=0.698132 rad]
D039	cross_resonator	9	1.000000	Steel-Rubber/Epoxy Cruciform Locally Resonant Phononic Crystal [L_arm/a=0.300000, h_arm/a=0.075000,

				t_halo/a=0.050000, theta=0.785398 rad]
D040	four_cylinder_cluster	0	0.000000	Tungsten-Rubber/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.055000, offset/a=0.130000, t_halo/a=0.018000]
D041	four_cylinder_cluster	1	0.111111	Steel-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.058333, offset/a=0.138333, t_halo/a=0.021111]
D042	four_cylinder_cluster	2	0.222222	Tungsten-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.061667, offset/a=0.146667, t_halo/a=0.024222]
D043	four_cylinder_cluster	3	0.333333	Steel-Rubber/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.065000, offset/a=0.155000, t_halo/a=0.027333]
D044	four_cylinder_cluster	4	0.444444	Tungsten-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.068333, offset/a=0.163333, t_halo/a=0.030444]
D045	four_cylinder_cluster	5	0.555556	Steel-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.071667, offset/a=0.171667, t_halo/a=0.033556]
D046	four_cylinder_cluster	6	0.666667	Tungsten-Rubber/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.075000, offset/a=0.180000, t_halo/a=0.036667]
D047	four_cylinder_cluster	7	0.777778	Steel-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.078333, offset/a=0.188333, t_halo/a=0.039778]
D048	four_cylinder_cluster	8	0.888889	Tungsten-Silicone/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.081667, offset/a=0.196667, t_halo/a=0.042889]
D049	four_cylinder_cluster	9	1.000000	Steel-Rubber/Epoxy Four-Inclusion Cluster Phononic Crystal [r_inc/a=0.085000, offset/a=0.205000, t_halo/a=0.046000]
D050	annular_resonator	0	0.000000	Tungsten-Rubber/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.110000, t_ring/a=0.050000, t_halo/a=0.020000]
D051	annular_resonator	1	0.111111	Steel-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.116667, t_ring/a=0.055000, t_halo/a=0.022778]
D052	annular_resonator	2	0.222222	Tungsten-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.123333, t_ring/a=0.060000, t_halo/a=0.025556]
D053	annular_resonator	3	0.333333	Steel-Rubber/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.130000,

				t_ring/a=0.065000, t_halo/a=0.028333]
D054	annular_resonator	4	0.444444	Tungsten-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.136667, t_ring/a=0.070000, t_halo/a=0.031111]
D055	annular_resonator	5	0.555556	Steel-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.143333, t_ring/a=0.075000, t_halo/a=0.033889]
D056	annular_resonator	6	0.666667	Tungsten-Rubber/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.150000, t_ring/a=0.080000, t_halo/a=0.036667]
D057	annular_resonator	7	0.777778	Steel-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.156667, t_ring/a=0.085000, t_halo/a=0.039444]
D058	annular_resonator	8	0.888889	Tungsten-Silicone/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.163333, t_ring/a=0.090000, t_halo/a=0.042222]
D059	annular_resonator	9	1.000000	Steel-Rubber/Epoxy Annular Ring-Resonator Phononic Crystal [r_inner/a=0.170000, t_ring/a=0.095000, t_halo/a=0.045000]
D060	split_ring_resonator	0	0.000000	Tungsten-Rubber/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.135000, h_ring/a=0.035000, gap_fraction=0.120000, theta=0.000000 rad]
D061	split_ring_resonator	1	0.111111	Steel-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.141111, h_ring/a=0.038333, gap_fraction=0.140000, theta=0.349066 rad]
D062	split_ring_resonator	2	0.222222	Tungsten-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.147222, h_ring/a=0.041667, gap_fraction=0.160000, theta=0.698132 rad]
D063	split_ring_resonator	3	0.333333	Steel-Rubber/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.153333, h_ring/a=0.045000, gap_fraction=0.180000, theta=1.047198 rad]
D064	split_ring_resonator	4	0.444444	Tungsten-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.159444, h_ring/a=0.048333, gap_fraction=0.200000, theta=1.396263 rad]
D065	split_ring_resonator	5	0.555556	Steel-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.165556, h_ring/a=0.051667, gap_fraction=0.220000, theta=1.745329 rad]
D066	split_ring_resonator	6	0.666667	Tungsten-Rubber/Epoxy Split-Ring Elastic Metamaterial

				[r_mean/a=0.171667, h_ring/a=0.055000, gap_fraction=0.240000, theta=2.094395 rad]
D067	split_ring_resonator	7	0.777778	Steel-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.177778, h_ring/a=0.058333, gap_fraction=0.260000, theta=2.443461 rad]
D068	split_ring_resonator	8	0.888889	Tungsten-Silicone/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.183889, h_ring/a=0.061667, gap_fraction=0.280000, theta=2.792527 rad]
D069	split_ring_resonator	9	1.000000	Steel-Rubber/Epoxy Split-Ring Elastic Metamaterial [r_mean/a=0.190000, h_ring/a=0.065000, gap_fraction=0.300000, theta=3.141593 rad]
D070	diamond_core_shell	0	0.000000	Tungsten-Rubber/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.135000, t_shell/a=0.035000, theta=0.000000 rad]
D071	diamond_core_shell	1	0.111111	Steel-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.145000, t_shell/a=0.040556, theta=0.087266 rad]
D072	diamond_core_shell	2	0.222222	Tungsten-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.155000, t_shell/a=0.046111, theta=0.174533 rad]
D073	diamond_core_shell	3	0.333333	Steel-Rubber/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.165000, t_shell/a=0.051667, theta=0.261799 rad]
D074	diamond_core_shell	4	0.444444	Tungsten-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.175000, t_shell/a=0.057222, theta=0.349066 rad]
D075	diamond_core_shell	5	0.555556	Steel-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.185000, t_shell/a=0.062778, theta=0.436332 rad]
D076	diamond_core_shell	6	0.666667	Tungsten-Rubber/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.195000, t_shell/a=0.068333, theta=0.523599 rad]
D077	diamond_core_shell	7	0.777778	Steel-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.205000, t_shell/a=0.073889, theta=0.610865 rad]
D078	diamond_core_shell	8	0.888889	Tungsten-Silicone/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.215000, t_shell/a=0.079444, theta=0.698132 rad]
D079	diamond_core_shell	9	1.000000	Steel-Rubber/Epoxy Rotated Diamond Core-Shell Phononic Crystal [R_L1/a=0.225000,

				t_shell/a=0.085000, theta=0.785398 rad]
D080	dimer_resonator	0	0.000000	Tungsten-Rubber/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.070000, offset/a=0.090000, t_halo/a=0.020000, theta=0.000000 rad]
D081	dimer_resonator	1	0.111111	Steel-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.073889, offset/a=0.100000, t_halo/a=0.023333, theta=0.174533 rad]
D082	dimer_resonator	2	0.222222	Tungsten-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.077778, offset/a=0.110000, t_halo/a=0.026667, theta=0.349066 rad]
D083	dimer_resonator	3	0.333333	Steel-Rubber/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.081667, offset/a=0.120000, t_halo/a=0.030000, theta=0.523599 rad]
D084	dimer_resonator	4	0.444444	Tungsten-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.085556, offset/a=0.130000, t_halo/a=0.033333, theta=0.698132 rad]
D085	dimer_resonator	5	0.555556	Steel-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.089444, offset/a=0.140000, t_halo/a=0.036667, theta=0.872665 rad]
D086	dimer_resonator	6	0.666667	Tungsten-Rubber/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.093333, offset/a=0.150000, t_halo/a=0.040000, theta=1.047198 rad]
D087	dimer_resonator	7	0.777778	Steel-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.097222, offset/a=0.160000, t_halo/a=0.043333, theta=1.221730 rad]
D088	dimer_resonator	8	0.888889	Tungsten-Silicone/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.101111, offset/a=0.170000, t_halo/a=0.046667, theta=1.396263 rad]
D089	dimer_resonator	9	1.000000	Steel-Rubber/Epoxy Dimerized Locally Resonant Phononic Crystal [r_disk/a=0.105000, offset/a=0.180000, t_halo/a=0.050000, theta=1.570796 rad]
D090	orthogonal_labyrinth	0	0.000000	Tungsten-Rubber/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.020000, inner_scale/a=0.110000, outer_scale/a=0.220000]
D091	orthogonal_labyrinth	1	0.111111	Steel-Silicone/Epoxy

				Labyrinthine Elastic Metamaterial [h_bar/a=0.022222, inner_scale/a=0.116667, outer_scale/a=0.228889]
D092	orthogonal_labyrinth	2	0.222222	Tungsten-Silicone/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.024444, inner_scale/a=0.123333, outer_scale/a=0.237778]
D093	orthogonal_labyrinth	3	0.333333	Steel-Rubber/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.026667, inner_scale/a=0.130000, outer_scale/a=0.246667]
D094	orthogonal_labyrinth	4	0.444444	Tungsten-Silicone/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.028889, inner_scale/a=0.136667, outer_scale/a=0.255556]
D095	orthogonal_labyrinth	5	0.555556	Steel-Silicone/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.031111, inner_scale/a=0.143333, outer_scale/a=0.264444]
D096	orthogonal_labyrinth	6	0.666667	Tungsten-Rubber/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.033333, inner_scale/a=0.150000, outer_scale/a=0.273333]
D097	orthogonal_labyrinth	7	0.777778	Steel-Silicone/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.035556, inner_scale/a=0.156667, outer_scale/a=0.282222]
D098	orthogonal_labyrinth	8	0.888889	Tungsten-Silicone/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.037778, inner_scale/a=0.163333, outer_scale/a=0.291111]
D099	orthogonal_labyrinth	9	1.000000	Steel-Rubber/Epoxy Labyrinthine Elastic Metamaterial [h_bar/a=0.040000, inner_scale/a=0.170000, outer_scale/a=0.300000]

Appendix K. Complete 100-Bulb Architecture Manifest

Exact finite-device names generated by the electromechanical bulb solver

The finite-bulb stage enumerated one hundred architectures. These are not one hundred copies of a single resonator; the family, material combination, dimensions, and piezoelectric assignment change across the manifest. D067 is shown in context rather than in isolation.

ID	Finite family	Piezo	Full engineering name
D000	Uniform Annular Core-Shell Resonator Bulb	PZT-5H	Tungsten-Silicone/Aluminum + PZT-5H Uniform Annular Core-Shell Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D001	Uniform Annular Core-Shell Resonator Bulb	PZT-5H	Steel-Silicone/Epoxy + PZT-5H Uniform Annular Core-Shell Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D002	Uniform Annular Core-Shell Resonator Bulb	PZT-5H	Brass-Silicone/Aluminum + PZT-5H Uniform Annular Core-Shell Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]

D003	Uniform Annular Core-Shell Resonator Bulb	PZT-5A	Tungsten-Silicone/Steel + PZT-5A Uniform Annular Core-Shell Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D004	Uniform Annular Core-Shell Resonator Bulb	PZT-5A	Tungsten-Silicone/Aluminum + PZT-5A Uniform Annular Core-Shell Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D005	Uniform Annular Core-Shell Resonator Bulb	PZT-5A	Steel-Silicone/Epoxy + PZT-5A Uniform Annular Core-Shell Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D006	Uniform Annular Core-Shell Resonator Bulb	AlN	Brass-Silicone/Aluminum + AlN Uniform Annular Core-Shell Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D007	Uniform Annular Core-Shell Resonator Bulb	AlN	Tungsten-Silicone/Steel + AlN Uniform Annular Core-Shell Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D008	Uniform Annular Core-Shell Resonator Bulb	AlN	Tungsten-Silicone/Aluminum + AlN Uniform Annular Core-Shell Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D009	Uniform Annular Core-Shell Resonator Bulb	LiNbO3	Steel-Silicone/Epoxy + LiNbO3 Uniform Annular Core-Shell Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D010	Dimerized Heavy-Light Resonator Bulb	PZT-5A	Brass-Rubber/Epoxy + PZT-5A Dimerized Heavy-Light Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D011	Dimerized Heavy-Light Resonator Bulb	PZT-5A	Tungsten-Rubber/Aluminum + PZT-5A Dimerized Heavy-Light Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D012	Dimerized Heavy-Light Resonator Bulb	PZT-5A	Tungsten-Rubber/Steel + PZT-5A Dimerized Heavy-Light Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D013	Dimerized Heavy-Light Resonator Bulb	AlN	Steel-Rubber/Aluminum + AlN Dimerized Heavy-Light Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D014	Dimerized Heavy-Light Resonator Bulb	AlN	Brass-Rubber/Epoxy + AlN Dimerized Heavy-Light Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D015	Dimerized Heavy-Light Resonator Bulb	AlN	Tungsten-Rubber/Aluminum + AlN Dimerized Heavy-Light Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D016	Dimerized Heavy-Light Resonator Bulb	LiNbO3	Tungsten-Rubber/Steel + LiNbO3 Dimerized Heavy-Light Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D017	Dimerized Heavy-Light Resonator Bulb	LiNbO3	Steel-Rubber/Aluminum + LiNbO3 Dimerized Heavy-Light Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D018	Dimerized Heavy-Light Resonator Bulb	LiNbO3	Brass-Rubber/Epoxy + LiNbO3 Dimerized Heavy-Light Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D019	Dimerized Heavy-Light Resonator Bulb	PZT-5H	Tungsten-Rubber/Aluminum + PZT-5H Dimerized Heavy-Light Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D020	Quadrupolar Cluster Resonator Bulb	AlN	Tungsten-Silicone/Aluminum + AlN

			Quadrupolar Cluster Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D021	Quadrupolar Cluster Resonator Bulb	AlN	Steel-Silicone/Steel + AlN Quadrupolar Cluster Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D022	Quadrupolar Cluster Resonator Bulb	AlN	Brass-Silicone/Aluminum + AlN Quadrupolar Cluster Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D023	Quadrupolar Cluster Resonator Bulb	LiNbO3	Tungsten-Silicone/Epoxy + LiNbO3 Quadrupolar Cluster Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D024	Quadrupolar Cluster Resonator Bulb	LiNbO3	Tungsten-Silicone/Aluminum + LiNbO3 Quadrupolar Cluster Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D025	Quadrupolar Cluster Resonator Bulb	LiNbO3	Steel-Silicone/Steel + LiNbO3 Quadrupolar Cluster Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D026	Quadrupolar Cluster Resonator Bulb	PZT-5H	Brass-Silicone/Aluminum + PZT-5H Quadrupolar Cluster Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D027	Quadrupolar Cluster Resonator Bulb	PZT-5H	Tungsten-Silicone/Epoxy + PZT-5H Quadrupolar Cluster Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D028	Quadrupolar Cluster Resonator Bulb	PZT-5H	Tungsten-Silicone/Aluminum + PZT-5H Quadrupolar Cluster Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D029	Quadrupolar Cluster Resonator Bulb	PZT-5A	Steel-Silicone/Steel + PZT-5A Quadrupolar Cluster Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D030	Split-Ring Defect Resonator Bulb	LiNbO3	Brass-Rubber/Steel + LiNbO3 Split-Ring Defect Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D031	Split-Ring Defect Resonator Bulb	LiNbO3	Tungsten-Rubber/Aluminum + LiNbO3 Split-Ring Defect Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D032	Split-Ring Defect Resonator Bulb	LiNbO3	Tungsten-Rubber/Epoxy + LiNbO3 Split-Ring Defect Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D033	Split-Ring Defect Resonator Bulb	PZT-5H	Steel-Rubber/Aluminum + PZT-5H Split-Ring Defect Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D034	Split-Ring Defect Resonator Bulb	PZT-5H	Brass-Rubber/Steel + PZT-5H Split-Ring Defect Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D035	Split-Ring Defect Resonator Bulb	PZT-5H	Tungsten-Rubber/Aluminum + PZT-5H Split-Ring Defect Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D036	Split-Ring Defect Resonator Bulb	PZT-5A	Tungsten-Rubber/Epoxy + PZT-5A Split-Ring Defect Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D037	Split-Ring Defect Resonator Bulb	PZT-5A	Steel-Rubber/Aluminum + PZT-5A Split-Ring Defect Resonator Bulb

			[R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D038	Split-Ring Defect Resonator Bulb	PZT-5A	Brass-Rubber/Steel + PZT-5A Split-Ring Defect Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D039	Split-Ring Defect Resonator Bulb	AlN	Tungsten-Rubber/Aluminum + AlN Split-Ring Defect Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D040	Elliptically Graded Resonator Bulb	PZT-5H	Tungsten-Silicone/Aluminum + PZT-5H Elliptically Graded Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D041	Elliptically Graded Resonator Bulb	PZT-5H	Steel-Silicone/Epoxy + PZT-5H Elliptically Graded Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D042	Elliptically Graded Resonator Bulb	PZT-5H	Brass-Silicone/Aluminum + PZT-5H Elliptically Graded Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D043	Elliptically Graded Resonator Bulb	PZT-5A	Tungsten-Silicone/Steel + PZT-5A Elliptically Graded Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D044	Elliptically Graded Resonator Bulb	PZT-5A	Tungsten-Silicone/Aluminum + PZT-5A Elliptically Graded Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D045	Elliptically Graded Resonator Bulb	PZT-5A	Steel-Silicone/Epoxy + PZT-5A Elliptically Graded Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D046	Elliptically Graded Resonator Bulb	AlN	Brass-Silicone/Aluminum + AlN Elliptically Graded Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D047	Elliptically Graded Resonator Bulb	AlN	Tungsten-Silicone/Steel + AlN Elliptically Graded Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D048	Elliptically Graded Resonator Bulb	AlN	Tungsten-Silicone/Aluminum + AlN Elliptically Graded Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D049	Elliptically Graded Resonator Bulb	LiNbO3	Steel-Silicone/Epoxy + LiNbO3 Elliptically Graded Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D050	Labyrinthine Compliant-Path Resonator Bulb	PZT-5A	Brass-Rubber/Epoxy + PZT-5A Labyrinthine Compliant-Path Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D051	Labyrinthine Compliant-Path Resonator Bulb	PZT-5A	Tungsten-Rubber/Aluminum + PZT-5A Labyrinthine Compliant-Path Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D052	Labyrinthine Compliant-Path Resonator Bulb	PZT-5A	Tungsten-Rubber/Steel + PZT-5A Labyrinthine Compliant-Path Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D053	Labyrinthine Compliant-Path Resonator Bulb	AlN	Steel-Rubber/Aluminum + AlN Labyrinthine Compliant-Path Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D054	Labyrinthine Compliant-Path Resonator Bulb	AlN	Brass-Rubber/Epoxy + AlN Labyrinthine Compliant-Path Resonator Bulb [R=31.111111mm, H=53.750000mm,

			t_shell=1.433333mm, coverage=0.594444]
D055	Labyrinthine Compliant-Path Resonator Bulb	AlN	Tungsten-Rubber/Aluminum + AlN Labyrinthine Compliant-Path Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D056	Labyrinthine Compliant-Path Resonator Bulb	LiNbO3	Tungsten-Rubber/Steel + LiNbO3 Labyrinthine Compliant-Path Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D057	Labyrinthine Compliant-Path Resonator Bulb	LiNbO3	Steel-Rubber/Aluminum + LiNbO3 Labyrinthine Compliant-Path Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D058	Labyrinthine Compliant-Path Resonator Bulb	LiNbO3	Brass-Rubber/Epoxy + LiNbO3 Labyrinthine Compliant-Path Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D059	Labyrinthine Compliant-Path Resonator Bulb	PZT-5H	Tungsten-Rubber/Aluminum + PZT-5H Labyrinthine Compliant-Path Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D060	Nested Inertial Core-Shell Resonator Bulb	AlN	Tungsten-Silicone/Aluminum + AlN Nested Inertial Core-Shell Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D061	Nested Inertial Core-Shell Resonator Bulb	AlN	Steel-Silicone/Steel + AlN Nested Inertial Core-Shell Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D062	Nested Inertial Core-Shell Resonator Bulb	AlN	Brass-Silicone/Aluminum + AlN Nested Inertial Core-Shell Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D063	Nested Inertial Core-Shell Resonator Bulb	LiNbO3	Tungsten-Silicone/Epoxy + LiNbO3 Nested Inertial Core-Shell Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D064	Nested Inertial Core-Shell Resonator Bulb	LiNbO3	Tungsten-Silicone/Aluminum + LiNbO3 Nested Inertial Core-Shell Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D065	Nested Inertial Core-Shell Resonator Bulb	LiNbO3	Steel-Silicone/Steel + LiNbO3 Nested Inertial Core-Shell Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D066	Nested Inertial Core-Shell Resonator Bulb	PZT-5H	Brass-Silicone/Aluminum + PZT-5H Nested Inertial Core-Shell Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D067	Nested Inertial Core-Shell Resonator Bulb	PZT-5H	Tungsten-Silicone/Epoxy + PZT-5H Nested Inertial Core-Shell Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D068	Nested Inertial Core-Shell Resonator Bulb	PZT-5H	Tungsten-Silicone/Aluminum + PZT-5H Nested Inertial Core-Shell Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D069	Nested Inertial Core-Shell Resonator Bulb	PZT-5A	Steel-Silicone/Steel + PZT-5A Nested Inertial Core-Shell Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D070	Fibonacci Binary-Mass Resonator Bulb	LiNbO3	Brass-Rubber/Steel + LiNbO3 Fibonacci Binary-Mass Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D071	Fibonacci Binary-Mass Resonator Bulb	LiNbO3	Tungsten-Rubber/Aluminum + LiNbO3 Fibonacci Binary-Mass Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]

D072	Fibonacci Binary-Mass Resonator Bulb	LiNbO3	Tungsten-Rubber/Epoxy + LiNbO3 Fibonacci Binary-Mass Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D073	Fibonacci Binary-Mass Resonator Bulb	PZT-5H	Steel-Rubber/Aluminum + PZT-5H Fibonacci Binary-Mass Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D074	Fibonacci Binary-Mass Resonator Bulb	PZT-5H	Brass-Rubber/Steel + PZT-5H Fibonacci Binary-Mass Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D075	Fibonacci Binary-Mass Resonator Bulb	PZT-5H	Tungsten-Rubber/Aluminum + PZT-5H Fibonacci Binary-Mass Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D076	Fibonacci Binary-Mass Resonator Bulb	PZT-5A	Tungsten-Rubber/Epoxy + PZT-5A Fibonacci Binary-Mass Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D077	Fibonacci Binary-Mass Resonator Bulb	PZT-5A	Steel-Rubber/Aluminum + PZT-5A Fibonacci Binary-Mass Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D078	Fibonacci Binary-Mass Resonator Bulb	PZT-5A	Brass-Rubber/Steel + PZT-5A Fibonacci Binary-Mass Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D079	Fibonacci Binary-Mass Resonator Bulb	AlN	Tungsten-Rubber/Aluminum + AlN Fibonacci Binary-Mass Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D080	Azimuthally Graded Resonator Bulb	PZT-5H	Tungsten-Silicone/Aluminum + PZT-5H Azimuthally Graded Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D081	Azimuthally Graded Resonator Bulb	PZT-5H	Steel-Silicone/Epoxy + PZT-5H Azimuthally Graded Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D082	Azimuthally Graded Resonator Bulb	PZT-5H	Brass-Silicone/Aluminum + PZT-5H Azimuthally Graded Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D083	Azimuthally Graded Resonator Bulb	PZT-5A	Tungsten-Silicone/Steel + PZT-5A Azimuthally Graded Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D084	Azimuthally Graded Resonator Bulb	PZT-5A	Tungsten-Silicone/Aluminum + PZT-5A Azimuthally Graded Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D085	Azimuthally Graded Resonator Bulb	PZT-5A	Steel-Silicone/Epoxy + PZT-5A Azimuthally Graded Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D086	Azimuthally Graded Resonator Bulb	AlN	Brass-Silicone/Aluminum + AlN Azimuthally Graded Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D087	Azimuthally Graded Resonator Bulb	AlN	Tungsten-Silicone/Steel + AlN Azimuthally Graded Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D088	Azimuthally Graded Resonator Bulb	AlN	Tungsten-Silicone/Aluminum + AlN Azimuthally Graded Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D089	Azimuthally Graded Resonator Bulb	LiNbO3	Steel-Silicone/Epoxy + LiNbO3

			Azimuthally Graded Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]
D090	Localized Defect-Cavity Resonator Bulb	PZT-5A	Brass-Rubber/Epoxy + PZT-5A Localized Defect-Cavity Resonator Bulb [R=28.000000mm, H=48.750000mm, t_shell=0.980000mm, coverage=0.350000]
D091	Localized Defect-Cavity Resonator Bulb	PZT-5A	Tungsten-Rubber/Aluminum + PZT-5A Localized Defect-Cavity Resonator Bulb [R=28.777778mm, H=50.000000mm, t_shell=1.093333mm, coverage=0.411111]
D092	Localized Defect-Cavity Resonator Bulb	PZT-5A	Tungsten-Rubber/Steel + PZT-5A Localized Defect-Cavity Resonator Bulb [R=29.555556mm, H=51.250000mm, t_shell=1.206667mm, coverage=0.472222]
D093	Localized Defect-Cavity Resonator Bulb	AlN	Steel-Rubber/Aluminum + AlN Localized Defect-Cavity Resonator Bulb [R=30.333333mm, H=52.500000mm, t_shell=1.320000mm, coverage=0.533333]
D094	Localized Defect-Cavity Resonator Bulb	AlN	Brass-Rubber/Epoxy + AlN Localized Defect-Cavity Resonator Bulb [R=31.111111mm, H=53.750000mm, t_shell=1.433333mm, coverage=0.594444]
D095	Localized Defect-Cavity Resonator Bulb	AlN	Tungsten-Rubber/Aluminum + AlN Localized Defect-Cavity Resonator Bulb [R=31.888889mm, H=55.000000mm, t_shell=1.546667mm, coverage=0.655556]
D096	Localized Defect-Cavity Resonator Bulb	LiNbO3	Tungsten-Rubber/Steel + LiNbO3 Localized Defect-Cavity Resonator Bulb [R=32.666667mm, H=56.250000mm, t_shell=1.660000mm, coverage=0.716667]
D097	Localized Defect-Cavity Resonator Bulb	LiNbO3	Steel-Rubber/Aluminum + LiNbO3 Localized Defect-Cavity Resonator Bulb [R=33.444444mm, H=57.500000mm, t_shell=1.773333mm, coverage=0.777778]
D098	Localized Defect-Cavity Resonator Bulb	LiNbO3	Brass-Rubber/Epoxy + LiNbO3 Localized Defect-Cavity Resonator Bulb [R=34.222222mm, H=58.750000mm, t_shell=1.886667mm, coverage=0.838889]
D099	Localized Defect-Cavity Resonator Bulb	PZT-5H	Tungsten-Rubber/Aluminum + PZT-5H Localized Defect-Cavity Resonator Bulb [R=35.000000mm, H=60.000000mm, t_shell=2.000000mm, coverage=0.900000]

Appendix L. Two-Degree-of-Freedom Audit Derivation

Analytic structure of the zero-dependency calculation

The zero-dependency audit intentionally reduces the device to a shell generalized coordinate X_s and a local resonator coordinate X_r. The purpose is not to reproduce every sector mode. The purpose is to expose the location of the principal resonances and the coupled load dependence in a formulation whose algebra can be solved without external numerical packages.

$$D11\ X_s + D12\ X_r - \Theta\ V = F_s$$
$$D21\ X_s + D22\ X_r = F_r$$
$$i\ \omega\ \Theta\ X_s + (1/R_L + i\ \omega\ C_p)\ V = 0$$

Eliminating the voltage gives an electrical back-action contribution to the shell dynamic stiffness. This term is complex and frequency dependent because the load has both dissipative and capacitive response.

$$V = -i\ \omega\ \Theta\ X_s / Y$$
$$D11,\mathrm{eff} = D11 + i\ \omega\ \Theta^2 / Y$$

The remaining 2x2 complex system is solved analytically. The undamped natural frequencies follow from det(K - lambda M) = 0 with lambda = omega^2. For the nominal audit the two branches were approximately 156.659049596398518 Hz and 780.045024052060285 Hz. The broadband search then recovered the lower branch as the useful nominal energy-conversion resonance.

L.1 Derivation detail

The base-excitation forcing is represented in relative coordinates, so generalized inertial forces scale with the effective masses and the imposed base acceleration. This makes displacement proportional to acceleration in the linear regime and power proportional to acceleration squared.

L.2 Derivation detail

Mechanical damping is represented by viscous terms for the shell and local relative motion. The dissipated mechanical power is computed from velocity amplitudes. Adding electrical load power gives a closed reduced-order input ledger.

L.3 Derivation detail

The shell and piezoelectric stresses used in the audit are generalized proxies derived from displacement divided by characteristic dimensions and multiplied by effective modulus. They are screening constraints, not substitutes for local tensor stresses from a finite-element model.

L.4 Derivation detail

The load search repeatedly converged near the expected RC scale. This is a useful implementation check because an optimum many orders of magnitude away from 1/(omega C_p) would require a clear physical explanation.

Appendix M. Energy, Power, and Efficiency Ledger

Definitions used in the reduced-order stages

Ledger quantity	Definition
Electrical load power	P_load = 0.5 |V|^2 / R_L
Rectified power	P_rect = eta_rect P_load
LED electrical power	P_LED,e = eta_DC P_rect
Optical power	P_opt = eta_LED P_LED,e
Mechanical damping loss	P_damp = 0.5 v* C v
Reduced input ledger	P_in = P_load + P_damp
Mechanical-to-electrical efficiency	eta_me = P_load / P_in
Total optical efficiency	eta_opt = P_opt / P_in

The input ledger is not a universal definition of total system input power. It is the power dissipated or extracted by the reduced harmonic model. A physical actuator may consume additional electrical power, and a real converter adds switching, conduction, control, and standby losses. The reported eta values therefore belong to the modeled electromechanical chain.

Reference piezo AC power: 1.29380171299124175e-04 W.

Reference optical power: 4.42790698254122647e-05 W.

Reference eta_me: 1.83361290460046060e-02.

Reference eta_opt: 6.27535680470461734e-03.

Peak voltage: 1.17064566155618577 V.

Appendix N. Full 32-Corner Manufacturing Matrix

Deterministic sign combinations used in tolerance screening

Corner	Mass +/-5%	k_c +/-5%	t_p +/-5%	Coverage +/-3%	Damping +/-10%
1	-1	-1	-1	-1	-1
2	-1	-1	-1	-1	+1
3	-1	-1	-1	+1	-1
4	-1	-1	-1	+1	+1
5	-1	-1	+1	-1	-1
6	-1	-1	+1	-1	+1
7	-1	-1	+1	+1	-1
8	-1	-1	+1	+1	+1
9	-1	+1	-1	-1	-1
10	-1	+1	-1	-1	+1
11	-1	+1	-1	+1	-1

12	-1	+1	-1	+1	+1
13	-1	+1	+1	-1	-1
14	-1	+1	+1	-1	+1
15	-1	+1	+1	+1	-1
16	-1	+1	+1	+1	+1
17	+1	-1	-1	-1	-1
18	+1	-1	-1	-1	+1
19	+1	-1	-1	+1	-1
20	+1	-1	-1	+1	+1
21	+1	-1	+1	-1	-1
22	+1	-1	+1	-1	+1
23	+1	-1	+1	+1	-1
24	+1	-1	+1	+1	+1
25	+1	+1	-1	-1	-1
26	+1	+1	-1	-1	+1
27	+1	+1	-1	+1	-1
28	+1	+1	-1	+1	+1
29	+1	+1	+1	-1	-1
30	+1	+1	+1	-1	+1
31	+1	+1	+1	+1	-1
32	+1	+1	+1	+1	+1

A + sign means the upper perturbation and a - sign means the lower perturbation. The set is deliberately symmetric and deterministic. It should be replaced by measured distributions and Monte Carlo yield calculations once prototype manufacturing statistics exist.

Appendix O. Passive Robustness Screen: Numerical Record

Selected fixed-operating-point results

The all-design fixed-point robustness screen evaluated every physical design at a single operating frequency and load across the 32 manufacturing corners. The best worst-case optical power plateaued early at 2.28401432071939221e-06 W and did not improve through the remainder of the 30000-design screen.

Rank	P_nom [W]	P_worst [W]	Retention	Damping scale
1	3.97047890825775703e-05	2.28401432071939221e-06	5.75249075361948178e-02	1.25
2	3.18689449370662708e-05	2.21105753367511056e-06	6.93796904177855084e-02	1.25
3	4.07301548464528645e-05	2.18956652344881986e-06	5.37578737842560953e-02	1.25
4	4.11901663659233606e-05	2.15726390403326935e-06	5.23732748459587311e-02	1.25
5	3.58036686466777491e-05	2.15297519566197603e-06	6.01328097661788727e-02	0.75
6	3.22586400266377624e-05	2.13600118719252310e-06	6.62148554752682583e-02	1.00
7	2.86358697139746455e-05	2.11469183463511597e-06	7.38476552574591910e-02	1.25
8	4.22155667545872460e-05	2.09831252419179113e-06	4.97047105014593524e-02	1.25
9	3.42111573782244744e-05	2.08763749907925720e-06	6.10221243321059370e-02	1.00
10	4.25114519952959685e-05	2.08378592487228770e-06	4.90170489848915369e-02	1.25

This record is sufficient to reject the idea that the nominal peak can simply be manufactured and operated at one immutable frequency and electrical load. The robust-design problem is fundamentally one of detuning management.

Appendix P. Adaptive Resonance-Tracking Control Law

A concrete implementation blueprint

The reduced-order study freezes the need for adaptation but does not freeze one controller. The following layered control architecture is recommended for the first prototype because each layer can be validated independently.

P.1 Fast measurement layer

- Acquire excitation reference, piezo voltage, converter input current, and optional structural response.
- Compute real electrical power over an integer number of mechanical cycles.
- Estimate response amplitude and phase relative to the excitation reference.

P.2 Resonance estimator

- Use phase crossing, amplitude extremum, or a local frequency perturbation to estimate the resonance offset.
- Reject transient estimates until the harmonic response settles sufficiently.
- Track resonance drift slowly compared with the carrier frequency.

P.3 Mechanical tuning

- Self-driven device: adjust actuator frequency.
- Externally excited device: adjust effective stiffness, inertial mass, or shunt state.
- Limit tuning rate to avoid destabilizing the mechanical oscillation.

P.4 Electrical tracking

- Estimate C_p or effective impedance at the current frequency.
- Select R_L or converter input impedance near the useful operating region.
- Optimize real extracted power, not open-circuit voltage.

P.5 Protection

- Enforce piezo stress proxy or measured strain limit.
- Enforce voltage, temperature, displacement, and actuator limits.
- Fall back to a safe detuned state when sensors disagree.

$$\text{objective} = \text{maximize } P\_\text{electrical} \text{ subject to stress, voltage, temperature, and actuator constraints}$$

Appendix Q. Prototype Test Record Definitions

Required fields for the first bench campaign; blank worksheet pages removed

The earlier edition included empty worksheet grids. They are replaced here by a compact data-schema definition so every table entry carries information. During experiments, the same fields can be implemented directly in CSV, HDF5, or a laboratory electronic notebook.

Campaign	Required recorded fields	Purpose
Modal sweep	Run ID; timestamp; temperature [K]; base acceleration [m/s^2]; frequency [Hz]; response amplitude; phase [rad or deg]; piezo open-circuit voltage [V]	Identify f_n, Q, modal amplitude, and phase response.
Load sweep	Run ID; frequency [Hz]; R_L [ohm]; V_rms [V]; I_rms [A]; real power [W]; temperature [K]; notes	Measure real electrical extraction and impedance dependence.
Detuning test	Perturbation ID; added mass/preload; initial f_r [Hz]; detuned f_r [Hz]; tracker recovery time [s]; recovered power [W]; retention ratio	Quantify adaptive resonance-tracking authority and recovery.
Thermal drift	Temperature [K]; f_r [Hz]; optimal R_L [ohm]; load power [W]; controller state; thermal dwell time [s]	Measure resonance and load drift with temperature.
Optical stage	Run ID; LED electrical input [W]; optical power [W]; wavelength/spectrum; detector calibration; ambient/background level	Close the electrical-to-optical ledger experimentally.
Safety / limits	Peak strain; inferred stress; peak displacement; peak voltage; component temperature; actuator command; trip state	Verify operation remains inside mechanical, electrical, and thermal limits.

Recommended file convention: one immutable raw-data file per run plus one derived-analysis file containing fitted resonance, Q, impedance optimum, uncertainty, and pass/fail flags. No result should overwrite the underlying raw measurement.

Appendix R. Publication and Peer-Review Audit Checklist

Questions that should be answered before external technical claims are strengthened

- Is every reported power explicitly labeled as modeled or measured?
- Are path band gaps distinguished from full-Brillouin-zone verified gaps?
- Are reduced-order stress proxies distinguished from local FEM stresses?
- Is the dependence of Theta and C_p on geometry physically linked?
- Are all optical efficiencies stated as assumptions unless measured?
- Is the controller required for robustness described as part of the architecture rather than an optional accessory?
- Are manufacturing tolerances stated and are fixed-frequency claims avoided?
- Is mesh convergence demonstrated for full-FEM results?
- Are experimental accelerations and electrical powers measured with calibrated instrumentation?
- Are negative or rank-reversing results preserved rather than omitted?
- Is the distinction between indicator-scale light and illumination-class output explicit?
- Are failure modes, fatigue, adhesives, thermal drift, and converter losses included before hardware performance claims?

Stopping rule: Do not extend the reduced-order parameter search merely to improve a number. Advance to full-field modeling or experiment once the architecture-level conclusion is stable and the remaining uncertainty is dominated by physics absent from the reduced-order model.

Appendix S. Engineering Glossary

Short definitions for cross-disciplinary readers

Term	Definition
Phononic crystal	A periodic elastic or acoustic structure whose dispersion can contain frequency regions with suppressed propagation.
Metacrystal	Here, a deliberately structured elastic unit cell or finite resonant architecture designed to manipulate mechanical-wave response.
Bloch condition	A periodic-wave boundary condition in which the field changes by a phase factor across a lattice translation.
Band gap	A frequency interval without propagating eigenmodes under the specified model and wave-vector domain.
Local resonance	A mode dominated by motion of an internal resonator relative to the host structure.
Piezoelectric coupling	Conversion between mechanical strain/stress and electric displacement/voltage.
Electrical back-action	Modification of mechanical response by the connected electrical load through piezoelectric coupling.
Quality factor Q	A measure of resonance sharpness; high Q generally means narrow bandwidth.
Adaptive resonance tracking	Closed-loop adjustment that keeps the device operating near its current resonant condition.
Worst-case corner	A deterministic combination of tolerance extremes used to screen robustness.
Reduced-order model	A compact model that preserves selected dynamic variables and couplings while omitting full spatial field resolution.
Architecture freeze	A decision that the system topology is stable enough to justify moving to a higher-fidelity validation stage.

Appendix T. D067-ART-v1 One-Page Technical Datasheet

Reduced-order reference configuration

Field	Reduced-order reference
Architecture	Nested inertial phononic resonator + PZT-5A + adaptive resonance tracking + adaptive load
Reference mode	Lower mechanical branch
Reference operating frequency	1.83592556468478932e+02 Hz
Reference load	5.29606372893091338e+03 ohm

Reference base acceleration	1.00000000000000000 m/s^2 peak
Piezo AC power	1.29380171299124175e-04 W
Modeled optical power	4.42790698254122647e-05 W
Peak voltage	1.17064566155618577 V
Shell safety factor	4.47813800228649495
Piezo safety factor	2.01643731830532191
Half-power bandwidth	3.06168876413914859 Hz
Effective Q	5.99644740572118309e+01
Passive fixed-frequency status	Rejected for manufacturing robustness
Control requirement	Adaptive resonance and electrical load tracking
Next validation stage	Finite-sector / 3-D piezoelectric FEM, then bench prototype

Final status: D067-ART-v1 is the frozen reduced-order architecture. It is a research design target for full-field simulation and experiment, not a certified product specification.